

THz WAVES OFFER BETTER IMAGING & SCREENING

THz radiation has good potential for application in medicine, security and defence. Unlike X-rays, it can penetrate deep without causing harmful photoionisation in biological tissues

■ ADITI SHARMA AND G. SINGH

Terahertz (THz) electromagnetic radiation has found increasing importance in the various fields of science and technology. Its frequency band spans from 100 GHz to 10 THz, and is often sited as the most scientifically rich regime of the electromagnetic spectrum. The former limit of the THz frequency spectrum lies just above the microwave region, whereas the latter limit is located adjacent to the infrared frequencies. Thus, the THz regime is a fusion between microwaves (electronics) and infrared light (photonics) as shown in Fig. 1. From the lower-frequency side, electronics keeps pushing upwards, while photonic approaches gradually improve technological options at higher frequencies.

Specifically, the THz regime offers important technical advantages compared to RF/microwave electronics and holds the promise for

novel sensing modalities. It has potential applications in imaging medicine, communication, homeland security and explosives' detection. Many more applications might emerge as the technology improves and matures.

The THz frequency range has led to unprecedented creativity in source development. It is mainly advances in the photonics branch, with the highlight being the development of THz quantum cascade lasers, which recently have caught the imagination of the public, potential users and investors.

Sources of high-quality THz radiation have been scarce, but this gap has recently begun to be filled by a wide range of new

powers up to 10 MW. New sources have led to new science in many areas, as scientists begin to become aware of the opportunities for research progress in their fields using THz radiation.

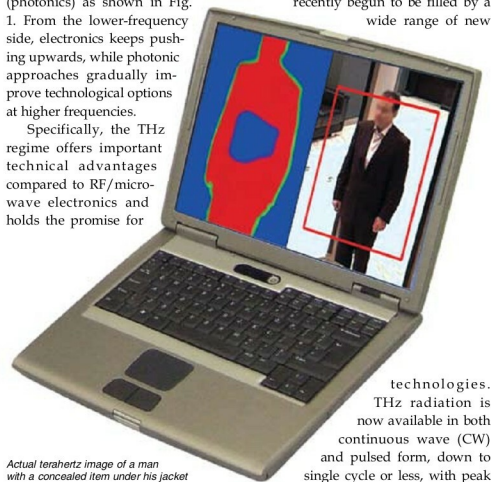
Potential applications

Fig. 2 shows the potential applications of THz electromagnetic radiation. THz radiation is widely used in bio-sensing. One of the most successful bio-sensors is currently based on surface plasmon resonance which relies on the refractive index change at optical wavelengths, as determined by the laser.

THz imaging and sensing. Sensing and imaging in the THz electromagnetic spectrum provides advantages for such security and defense applications as screening of personnel for hidden objects and detection of chemical and biological agents. As a consequence of optical diffraction, one can achieve higher-resolution imagery in the THz band with order-of-magnitude smaller imaging apertures. This makes THz imaging systems a practical option in the field of operations for many military platforms and missions.

THz imaging can achieve ten times better depth of field than passive millimetre waves and phase information—imaging radar—can be extracted at THz frequencies. A THz imager could enable a tank traveling through dust and fog to see the vehicle ahead of it.

Time-domain THz spectroscopy, with its time coherence and extremely broad spectral bandwidth, will continue to expand its reach and range of applications, from spectroscopy of superconductors to subcutaneous



Actual terahertz image of a man with a concealed item under his jacket

technologies. THz radiation is now available in both continuous wave (CW) and pulsed form, down to single cycle or less, with peak